Worksheet 5-7 & 5-12 A
2nd Derivative Test and Implicit Functions #1
AP Calculus

Name SOLUTIONS

NO CALCULATORS

x	0	1	4	6	9
$f(x)$	7	12	9	2	5
$f'(x)$	2	0	-3	0	1
$f''(x)$	3	1	0	-2	0

1. Let $f(x)$ be a twice differentiable function with selected values of f, f' and f'' given in the table above. For which x value(s) does $f(x)$ have a relative minimum? Give a reason for your answer.

$f(x)$ has a relative minimum at $x=1$ because $f'(1)=0$ and $f''(1)=1>0$

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2. Consider the function $y = f(x)$ where $\frac{dy}{dx} = \frac{8-2x}{3y^2}$. It is known that $f(4) = 3$.

A) Find $\frac{d^2y}{dx^2}$.

$$\frac{d^2y}{dx^2} = \frac{3y^2(-2) - (8-2x)\left(6y\frac{dy}{dx}\right)}{(3y^2)^2}$$

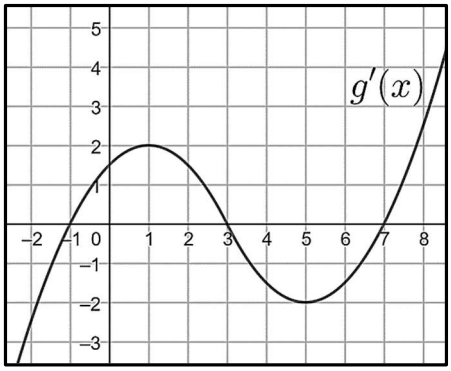
$$= \frac{-6y^2 - (8-2x)(6y)\left(\frac{8-2x}{3y^2}\right)}{(3y^2)^2}$$

OR $$\frac{-18y^4 - 6y(8-2x)^2}{(3y^2)^3}$$

B) Does $f(x)$ have a relative minimum, relative maximum, or neither at the point (4,3)? Explain your reasoning.

$$\left.\frac{d^2y}{dx^2}\right|_{(4,3)} = \frac{-6(9) - (8-8)(18)\left(\frac{8-8}{27}\right)}{27^2} = \frac{-54}{27^2} < 0$$

There is a relative maximum at the point (4,3) because $\frac{dy}{dx}=0$ and $\frac{d^2y}{dx^2}<0$.



x	-1	0	2	3	7
$h(x)$	-4	3	0	-2	0
$h'(x)$	6	-7	1	0	5

$f''(x) = g''(x)h(x) + g'(x) \cdot h'(x)$

3. Let f be a twice differentiable function where $f'(x) = g'(x) \cdot h(x)$. The graph of $g'(x)$ is shown in the figure above along with a table of $h(x)$ and $h'(x)$ at selected values of x.

For each of the following values of x, determine if the graph of $f(x)$ has a local minimum, local maximum, or neither. If it is not possible to determine if $f(x)$ has a local extremum, write "cannot be determined". Give a reason for your answers.

A) $x = -1$

$f'(-1) = g'(-1) \cdot h(-1)$
$= 0(-4) = 0$

$f''(-1) = g''(-1)h(-1) + g'(-1)h'(-1)$
$\approx (2)(-4) + 0(6) < 0$

$f(x)$ has a local maximum at $x = -1$ because $f'(-1) = 0$ and $f''(-1) < 0$.

B) $x = 0$

$f'(0) = g'(0)h(0)$
$\approx 1.6(3)$.

$f(x)$ does not have a local extremum at $x = 0$ because $f'(0) \neq 0$.

C) $x = 3$

$f'(3) = g'(3)h(3)$
$= 0(-2) = 0$

$f''(3) = g''(3)h(3) + g'(3)h'(3)$
$\approx (-2)(-2) + 0(0) > 0$

$f(x)$ has a local minimum at $x = 3$ because $f'(3) = 0$ and $f''(3) > 0$.

D) $x = 7$

$f'(7) = g'(7)h(7)$
$= 0(0) = 0$

$f''(7) = g''(7)h(7) + g'(7)h'(7)$
$\approx 2(0) + 0(0) = 0$

It cannot be determined because $f'(7) = 0$ and $f''(7) = 0$

4. The function $y = k(x)$ is continuous for all real values of x and has the differential equation $\frac{dy}{dx} = \frac{2x+3}{3(y+1)^2}$. It is known that $k(2) = -1$.

A) At which value(s) of x does the graph of k have a vertical tangent line?

$\frac{dy}{dx}$ is undefined when $3(y+1)^2 = 0$

$y+1 = 0$

$y = -1 \rightarrow x = 2$

$\left.\frac{dy}{dx}\right|_{(2,-1)} = \frac{4+3}{0}$

$=$ undef.

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5. For $x > 0$, the function $y = f(x)$ is continuous with $\frac{dy}{dx} = \frac{y}{x} + 1$. It is known that $f(1) = -1$.

A) Write an equation of the line tangent to the graph of f at $x = 1$.

slope $= \left.\frac{dy}{dx}\right|_{(1,-1)} = \frac{-1}{1} + 1 = -1 + 1 = 0$

EQ_{TAN}: $y + 1 = 0(x-1)$

OR $y = -1$

B) Find an expression for $\frac{d^2y}{dx^2}$ in terms of x and y.

$$\frac{d^2y}{dx^2} = \frac{x \cdot \frac{dy}{dx} - y}{x^2} = \frac{x\left(\frac{y}{x} + 1\right) - y}{x^2} = \frac{y + x - y}{x^2} = \frac{x}{x^2} = \frac{1}{x}$$

C) What is the value of $\frac{d^2y}{dx^2}$ when $x = 1$? Does f have a local minimum, local maximum, or neither at $x = 1$? Give a reason for your answer.

$\left.\frac{dy}{dx}\right|_{(1,-1)} = 0$ $\left.\frac{d^2y}{dx^2}\right|_{x=1} = \frac{1}{1} = 1 > 0$

f has a local minimum at $f(1) = -1$ because $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$.

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6. Consider the curve defined by $\frac{dy}{dx} = y \sin x$ where $f(2\pi) = 1$. Does f have a local minimum, local maximum, or neither at the given point? Give a reason for your answer.

$\left.\frac{dy}{dx}\right|_{(2\pi,1)} = 1 \sin 2\pi = 0$

$\frac{d^2y}{dx^2} = \frac{dy}{dx} \cdot \sin x + y \cdot \cos x$

$\left.\frac{d^2y}{dx_2}\right|_{(2\pi,1)} = 0 \cdot \sin 2\pi + 1 \cdot \cos 2\pi$

$= 1 > 0$

f has a local minimum at $f(2\pi) = 1$ because $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} > 0$.

7. Consider the curve defined by $3x^2 - xy + 2y = -36$ with $\frac{dy}{dx} = \frac{y - 6x}{2 - x}$.

A) Show that there are no points where the line tangent to the curve is vertical.

$\frac{dy}{dx}$ is undefined when $2-x=0$

$x=2$

$3(2)^2 - 2y + 2y = -36$

$12 = -36$

not possible.

There are no points on the curve where $x=2$.

B) Find the two points where the curve has a horizontal tangent line.

$\frac{dy}{dx} = 0$

when $y - 6x = 0$

$y = 6x$

$3x^2 - x(6x) + 2(6x) = -36$

$3x^2 - 6x^2 + 12x = -36$

$-3x^2 + 12x + 36 = 0$

$-3(x^2 - 4x - 12) = 0$

$-3(x+2)(x-6)$

$x = -2 \quad x = 6$

$x=-2$: $12 + 2y + 2y = -36$

$12 + 4y = -36$

$4y = -48$

$y = -12$

$x = 6$: $3(36) - 6y + 2y = -36$

$108 - 4y = -36$

$-4y = -144$

$y = 36$

horizontal tangent lines are at $(-2,-12)$ and $(6,36)$

C) Write an equation of the line tangent to the curve when $x = 0$.

$3x^2 - xy + 2y = -36$

$0 - 0 + 2y = -36$

$2y = -36$

$y = -18$

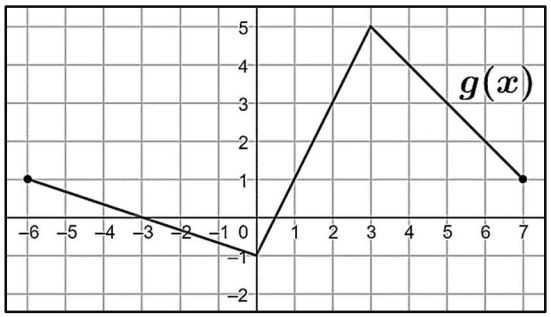
$\left.\frac{dy}{dx}\right|_{(0,-18)} = \frac{-18 - 6(0)}{2-0} = -\frac{18}{2} = -9$

EQ_{TAN}: $y + 18 = -9(x - 0)$

OR

$y + 18 = -9x$

$y = -9x - 18$



x	0	1	2	3	5
$h'(x)$	0	-6	0	2	0
$h''(x)$	4	-5	-3	0	-1

The function $g(x)$ is continuous on the interval $[-6, 7]$ and consists of three line segments as shown in the figure above. The function $h(x)$ is twice differentiable with $h(x) < 0$ for all real values of x. Selected values for h' and h'', the derivatives of h, are included in the table above.

8. Let $f(x)$ be a function where $f'(x) = g(1 - 2x)$. Does f have a local minimum, local maximum, or neither at $x = 2$? Give a reason for your answer.

$f'(2) = g(1-4) = g(-3) = 0$

$f''(x) = g'(1-2x)(-2)$

$f''(2) = g'(-3)(-2)$

$= -\frac{1}{3}(-2) = \frac{2}{3} > 0$

f has a local minimum at $x=2$ because $f'(2)=0$ and $f''(2)>0$.

9. Let $p(x)$ be a function where $p'(x) = g(x^2 + x)$. Does p have a local minimum, local maximum, or neither at $x = -1$? Give a reason for your answer.

$p'(-1) = g(1-1) = g(0) = -1$

P does not have a local max or local minimum at $x=-1$ because $p'(-1) \neq 0$

10. Does the function h have a relative minimum, relative maximum, or neither at $x = 5$? Explain your reasoning.

$h'(5)=0$

$h''(5)=-1<0$

h has a relative maximum at $x=5$ because $h'(5)=0$ and $h''(5)<0$.

11. Does the function h have a relative minimum, relative maximum, or neither at $x = 3$? Explain your reasoning.

$h'(3)=2$

h does not have a relative maximum nor relative minimum at $x=3$ because $h'(3) \neq 0$.